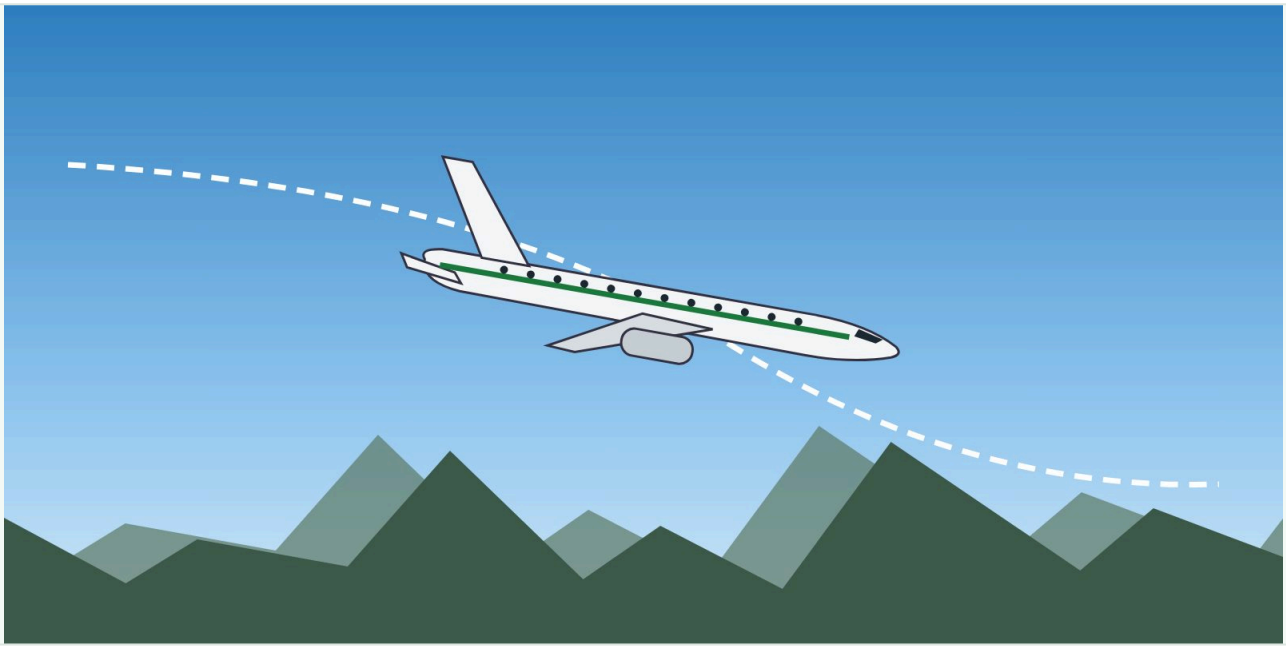
MODULE 3 · CERTIFICATION, PERFORMANCE AND LOADING

Lesson 10: Energy Management and Pitch-Power Coupling

Lesson 10 of 24 · about 8 min

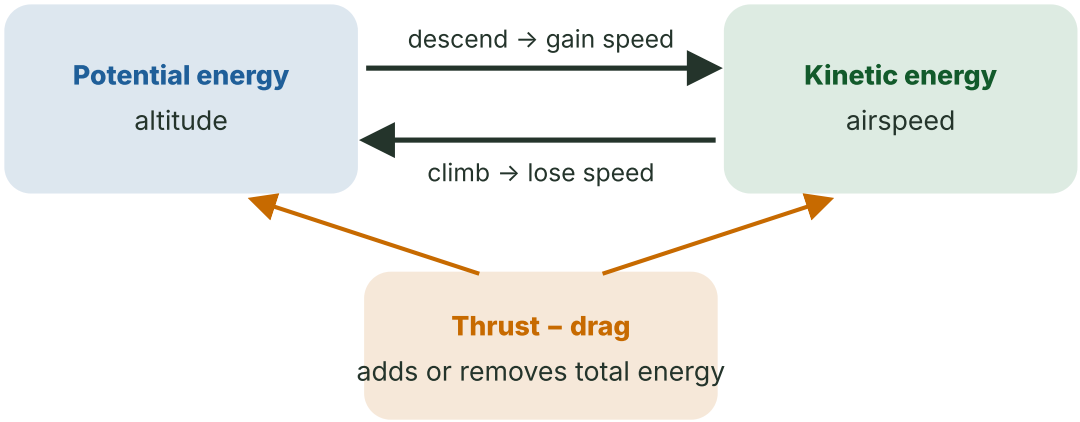
LEARNING OBJECTIVES

- Describe aircraft energy states and how pitch, thrust and drag change them.
- Explain the effect of underwing engines on pitch.
- Apply energy awareness to prevention and recovery.



Altitude and speed are two forms of the same energy.

Energy states



Total energy is altitude plus speed. Pitch exchanges one for the other; only thrust and drag change the total. Know which state you are in before you act.

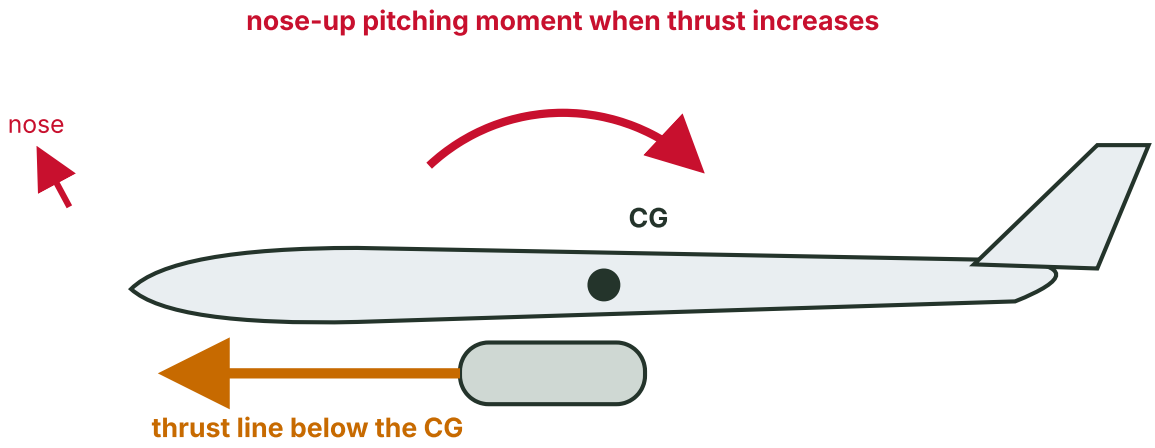
An aircraft's total energy is the sum of its **potential energy** (altitude) and **kinetic energy** (airspeed). Pitch exchanges one for the other. Only **thrust and drag** change the total. Common unsafe energy states:

- **Low energy** — low speed and low altitude, for example after a speed decay on approach. Few options remain.
- **High energy** — high speed in a descent, for example in a nose-low upset. The risk is overspeed and structural overload.
- **Mismatch** — high pitch with decaying speed (the energy is going into the climb), or low thrust with a level attitude.

Thrust response

- Jet engines take several seconds to accelerate from idle. Thrust is not an instant cure for a low-energy state — anticipate.
- At high altitude, excess thrust is small (Lesson 7).
- On the "back side" of the drag curve, below minimum drag speed, slowing down *increases* drag and the thrust needed. Speed can decay quickly if not corrected.

Pitch-power coupling



With engines mounted below the wing, adding thrust pitches the nose up and reducing thrust pitches it down. At low speed, with little elevator authority, this coupling can be strong.

With engines mounted below the wing, the thrust line is below the centre of gravity. Increasing thrust produces a **nose-up** pitching moment; reducing thrust produces a nose-down moment. Expect it during a go-around, and in recoveries:

- In a **nose-high, low-speed** situation, high thrust can make it harder to lower the nose. Reducing thrust may be necessary to regain pitch control.
- In a **stall recovery**, adding thrust too early can work against the nose-down input.
- A go-around with full thrust from a low speed produces a strong pitch-up; follow the pitch target and trim as required.

Speedbrakes and configuration

Speedbrakes increase drag and, on many types, the stall speed. They are a tool for high-energy situations — such as a nose-low, high-speed upset — but should be retracted in a low-energy situation or stall recovery. Do not change the flap or gear configuration in a recovery unless the procedure requires it.

KEY TAKEAWAYS

- Total energy = altitude + speed. Pitch exchanges them; thrust and drag change the total.
- Jet thrust takes time to build — anticipate low-energy situations.
- Underwing engines: more thrust = nose-up moment; less thrust = nose-down.
- Speedbrakes for high energy only; retract them in a stall recovery.

KNOWLEDGE CHECK

In a nose-high, low-speed situation on an aircraft with underwing engines, why may thrust need to be reduced?

A. To save fuel

B. Because high thrust creates a nose-up moment that resists lowering the nose

C. To increase the stall speed

D. To engage the autopilot

Answer: B. The thrust line below the CG produces a nose-up moment. Reducing thrust helps the elevator lower the nose.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.